

Arjun Thanneermalai Shanker

arjun70k.g@gmail.com | +60-16-421-9121 | Malaysian | 11900 Bayan Lepas, Pulau Pinang
github.com/Thaneerrjun7 | thaneerrjun7.github.io/Arjun-Portfolio | linkedin.com/in/arjun-shanker-ba25b3227
Software Engineer | Python, LLM Applications and Machine Learning | Testing and CI/CD

SUMMARY

Software Engineering graduate (dual degree, Sunway and Lancaster Universities) with a four-month internship, available immediately and open to relocate. In Python I built the large language model (LLM) layer of KIRA+'s AI financial assistant and trained a machine learning classifier, and for two live government portals I built a Flutter front end and a Figma design system. I learn quickly, work well independantly and in a team, and wish to learn & work on AI/Machine Learning for Real Software.

EDUCATION

Lancaster University and Sunway University

Bandar Sunway, Selangor

Bachelor of Software Engineering (Hons), Dual Degree (Twinning Program)

Sep 2023 - Sep 2026

CGPA 3.26 / 4.00 | Dean's List: Feb - Apr 2025 (GPA 3.67), May - Aug 2026 (GPA 3.63) | Student Ambassador (2024, 2025)

- Related Coursework: Data Structures and Algorithms, Software Architecture and Design Patterns (A+), Software Processes (A+, DevOps and CI/CD), Concurrent Programming (Java multithreading and thread pools, CUDA labs), Operating Systems (C, Linux, processes and threads), Artificial Intelligence, Object-Oriented Programming (Java, C++), Computer Vision.

Sunway College, OSSD (CIMP), Canadian International Matriculation Programme

Jul 2022 - Jul 2023

EXPERIENCE

Century Software (Malaysia) Sdn Bhd

Petaling Jaya, Selangor

Software Engineer Intern

Jan 2026 - Apr 2026

- Implemented the Flutter front end of two live government portals for Android and iOS as a component library in code, wired it to their existing back-end APIs after analysing the database schemas, and ran pre-release and regression tests.
- As the team's only UI/UX designer, designed 500+ screens for BayarNow (bill payments for the public across Malaysia) and OSC (business licensing), on a Figma component library self-built to the Malaysian Government Design System (MYDS).
- Proposed Pay Direct, a no-login payment flow adopted over the original plan, and wrote its storyboard. Collaborated with the CEO, CFO and product manager and supervisor on requirements and presented prototypes to the board of directors.

Fireball (fireball.my)

Remote

Freelance Website Developer

Nov 2025 - Feb 2026

- Alongside full-time studies, co-built the official Malaysian website of a Korean automotive accessories brand, linking its products to the client's Shopee store for checkout. Handled all client communication and took every post-launch issue (Shopee payment problems, the site not loading) through to a fix.

PROJECTS

KIRA+: AI Financial Health Platform for Buy Now, Pay Later and Consumer Credit

2026, in progress

MAIC Nexus Challenge 2026, Fintech track | AI/ML developer role, distributed team of 5 | Python, FastAPI, SQLite, GitHub Actions

- Built its AI financial assistant's LLM layer, the system prompt, a provider-switching client that answers within 4 seconds, and an output guard that rejects any reply quoting a figure the engine did not compute and prevents giving financial advice.
- Implemented the 0-100 scoring engine (six weighted risk factors, six warning rules) and trained and evaluated a RandomForest classifier on the project's 12,000 synthetic profiles: ROC-AUC 0.923 held out, 0.921 in 5-fold cross-validation.
- Worked test-first (56 unit and integration tests), then ran the app in a browser and fixed two defects the suites missed. Built the Docker image and GitHub Actions CI/CD pipeline with a teammate, and provisioned the server it deploys to.

SivaSafe: Voice-Controlled Desktop Assistant with Facial Authentication for Elderly Use

2025 - 2026

Final Year Capstone Project, graded A | Independent project over two semesters | Python end to end, PyQt6, OpenCV, Gemini

- Designed the assistant around one Gemini session that holds the spoken conversation and calls tools in 19 capability areas, choosing one agent over the proposal's multi-agent design to cut cost and delay.
- Secured sign-in with face recognition (OpenCV YuNet detection, SFace embeddings matched by cosine similarity), a spoken password transcribed offline by a Whisper model from Hugging Face, and PBKDF2-hashed recovery codes.
- Found and fixed an API-key failover that never ran (it watched for the wrong error type), root-caused two race conditions in its messaging automation, and measured Gemini's response time at about 1.1 s for a short reply and 5.2 s for reasoning.

Local AI Assistant: Tool-Calling AI Agent

2025, personal project

- Defined the tools a Gemini 2.5 Flash agent uses to control the computer by voice and wrote the code behind each, then chained calls, feeding each result back until the request was done, with fallback across Gemini, Groq and OpenRouter.

Portfolio Site: hand-written HTML, CSS and JavaScript on GitHub Pages, with its content served live from Cloud Firestore

TECHNICAL SKILLS

Programming: Python 3.14, Java, C++, C, SQL, JavaScript, TypeScript, Dart, HTML, CSS, PowerShell, Bash

AI and Machine Learning: LLM APIs (Gemini, OpenAI, Anthropic, Groq, OpenRouter), prompt engineering, output guardrails, tool-calling AI agents, LangChain (DeepLearning.AI), Hugging Face, OpenCV, scikit-learn, pandas, NumPy

Back End and Front End: FastAPI, REST APIs, Node.js, SQLite, Cloud Firestore (NoSQL), Flutter, npm, Figma

DevOps and Tools: Git version control, GitHub Actions CI/CD, Docker (Dockerfile, CI builds), DigitalOcean, Linux over SSH

Testing and Practice: pytest, unit, integration and regression tests, test-driven development, debugging, Agile and Scrum

CERTIFICATIONS AND LANGUAGES

Microsoft Certified: Azure AI Fundamentals (AI-900), 2025 | Microsoft AI-102 Exam-Prep Training, Apr - Aug 2025

LangChain for LLM Application Development (DeepLearning.AI), 2026

Languages: English (Fluent), Malay (Fluent), Tamil (Native), Mandarin Chinese (Basic, currently studying)

18 May 2026

TO WHOM IT MAY CONCERN

RE: Recommendation for Mr. Arjun Thanneermalai A/L Shanker

This letter serves as a recommendation for Mr. Arjun Thanneermalai A/L Shanker, who successfully completed his three-and-a-half-month internship with Century Software (M) Sdn Bhd.

During his internship, Arjun was assigned to design the UI/UX for our One-Stop Center (OSC) mobile application project using Figma. Throughout the internship period, he consistently demonstrated professionalism, creativity, and a strong sense of responsibility in carrying out his tasks.

Arjun was able to work independently with minimal supervision and showed great initiative in understanding the system requirements and producing relevant and practical design solutions. He is a fast learner who adapts quickly to new concepts and challenges. His dedication, commitment, and positive working attitude enabled him to consistently deliver quality designs across various modules and processes of the mobile application.

In addition to his technical and creative abilities, Arjun also displayed good communication skills and professionalism in his interactions with the team. He proved himself to be reliable, hardworking, and eager to learn throughout his internship with us.

We believe Arjun has strong potential and would be a valuable addition to any organization. We would not hesitate to recommend him for future employment opportunities.

Yours sincerely,



Alvin Mah
Development - Head
Century Software (M) Sdn Bhd



Ontario

Ontario Secondary School Diploma Diplôme d'études secondaires de l'Ontario

This Diploma is granted to
Ce diplôme est décerné à

ARJUN THANNEERMALAI A/L SHANKER

a student of
élève de

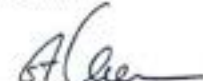
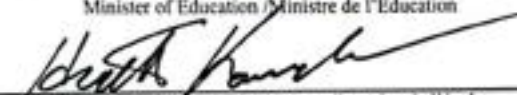
SUNWAY COLLEGE, MALAYSIA

who has fulfilled the requirements for the Ontario Secondary School Diploma
in accordance with the provisions of the Ministry of Education, Ontario.

qui a rempli les exigences prescrites pour l'obtention du diplôme d'études secondaires de l'Ontario,
en vertu des dispositions du ministère de l'Éducation de l'Ontario.

Dated at
Délivré à Bandar Sunway, Malaysia

the 16th day of June 2023
ce jour de


Minister of Education / Ministre de l'Éducation

Principal of School / Directeur ou directrice de l'école

SUNWAY
UNIVERSITY



Dean's List Award

is awarded to

***ARJUN THANNEERMALAI A/L
SHANKER***

in recognition of outstanding academic performance

for the February 2025 Semester in

Bachelor of Software Engineering (Hons)

A handwritten signature in black ink, appearing to read 'Denny Ng Kok Sum'.

Prof. Ir Denny Ng Kok Sum
Professor and Dean
Faculty of Engineering and Technology

24 April 2025



ARJUN THANNEERMALAI A/L SHANKER

has successfully passed all requirements for

Microsoft Certified: Azure AI Fundamentals

Credential ID: 267DA8B62CBC45D4

Certification number: 70YCDF-85A2BC

Earned on: July 27, 2025

✓ Online Verifiable



Satya Narayana Nadella

16 June 2023

Dear Graduate,

Sincere congratulations on your graduation from the Canadian International Matriculation Programme (CIMP). As you complete this stage of your educational journey, we anticipate that the Ontario Secondary School Diploma which you received today will be your Passport to Success.

In the years to come, we hope you will reflect fondly on the happy memories of your time with us. We are pleased that you chose CIMP as your Pre-university programme. We trust that your parents, your friends and you are as proud of your accomplishments as we are. We are all united today by a shared sense of delight in your achievements.

On behalf of CIMP and Sunway College, we wish you a future filled with the benefits of your achievements and the joys of lifelong learning.

Sincerely,



Mr. Heath Kondro

Director

Canadian International Matriculation Programme
Sunway College Kuala Lumpur

Sunway College (Kuala Lumpur) (DK25-01/9)
Sunway College (KL) Sdn Bhd 200401015440 (033945-A)
A member of the Sunway Education Group

No. 2, Jalan Universiti,
Bandar Sunway,
47500 Selangor Darul Ehsan
Tel: +6 03 5638 7176
college.sunway.edu.my

V01-06-01, Lingkaran SV,
Sunway Velocity,
55100 Kuala Lumpur
Tel: +6 03 9770 1155
college.sunway.edu.my/velocity



Owned and governed by the

Jeffrey Cheah
Foundation 
Nurturing the Seeds of Wisdom



On behalf of the Government of Ontario, we are very pleased to recognize your tremendous achievement and hard work.

As you graduate and embark on your next journey, know that your community and province are proud of you.

As you take the next steps in your life – whether through university, college, the skilled trades – we are confident you have the skills and ability to shine bright.

Bring with you the values that make Ontario's students the best in the world — a strong sense of duty, respect for differences, and determination to succeed.

Wishing you great success in the year ahead.

Au nom du gouvernement de l'Ontario, nous vous félicitons pour vos remarquables accomplissements et votre travail assidu.

Alors que vous entamez l'étape qui suit l'obtention de votre diplôme, sachez que votre communauté et votre province sont fières de vous.

Nous sommes convaincus que vous avez acquis les compétences et la capacité nécessaires pour exceller dans l'itinéraire que vous avez choisi, qu'il s'agisse de l'université, du collège ou des métiers spécialisés.

Soyez toujours fidèles aux valeurs qui font des élèves de l'Ontario les meilleurs au monde : un sens profond du devoir, le respect des différences et la détermination de réussir.

Tous nos vœux de réussite pour l'année qui vous attend.

The Honourable Doug Ford
Premier of Ontario

L'honorable Doug Ford
Premier ministre de l'Ontario

The Honourable Stephen Lecce
Minister of Education

L'honorable Stephen Lecce
Ministre de l'Éducation



Surname SHANKER		Given Names ARJUN THANNEERMALAI		OEN/MIN 680-689-668		Student Number 0A2270154		Date of Birth Year: 2002 Month: 10 Day: 02			
Name of District School Board / School Authority OFFSHORE SCHOOL BOARD			Number BOARD # 78000		Name of School SUNWAY COLLEGE			Number 879045		Date of Entry Year: 2022 Month: 07 Day: 27	

Date Year Month	Course Grade/Level	Course Title	Course Code	Percentage Grade	Credit	Compulsory	Note
2022-07		Equivalent Credits	PLE	EQV	24.00	17.00	
2022-12	12	English	ENG4U	61	1.00	X	
2022-12	12	Advanced Functions	MHF4U	65	1.00		
2022-12	12	Chemistry	SCH4U	67	R		
2023-06	12	Calculus and Vectors	MCV4U	64	1.00		
2023-06	12	Financial Accounting Principles	BAT4M	86	1.00		
2023-06	12	Ontario Secondary School Literacy Course	OLC4O	71	1.00		
2023-06	12	Biology	SBI4U	67	1.00		
2023-06	12	Chemistry	SCH4U	70	1.00		
SUMMARY OF CREDITS						31.00	18.00

Community Involvement ☒ Completed ☐ N/A	Provincial Secondary School Literacy Requirement ☒ Successfully Completed ☐ N/A	Secondary School Online Learning Requirement ☐ Successfully Completed ☒ N/A	Specialized Program
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Diploma or Certificate Ontario Secondary School Diploma	Date of Issue Year: 2023 Month: 06	Authorization Mr. Heath Kondro
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The collection and maintenance of this information are authorized under the Education Act, R.S.O., 1990, c. E.2, s. 286. Users of this information are supervisory officers and the principal and teachers of the school. This is the official record of the student's educational history. The contact person regarding the collection of this information is the principal of the school.



Ontario

Ontario Secondary School Diploma Diplôme d'études secondaires de l'Ontario

This Diploma is granted to

Ce diplôme est décerné à

ARJUN THANNEERMALAI A/L SHANKER

a student of

élève de

SUNWAY COLLEGE, MALAYSIA

who has fulfilled the requirements for the Ontario Secondary School Diploma

in accordance with the provisions of the Ministry of Education, Ontario.

qui a rempli les exigences prescrites pour l'obtention du diplôme d'études secondaires de l'Ontario.

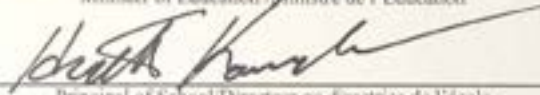
en vertu des dispositions du ministère de l'Éducation de l'Ontario.

Dated at
Délivré à

Bandar Sunway, Malaysia

the 16th day of June 2023
ce 16^e jour de


Minister of Education / Ministre de l'Éducation


Principal of School / Directeur ou directrice de l'école

Sekolah Menengah Kebangsaan St. George, Taiping.



SURAT AKUAN

TESTIMONIAL

Kepada Sesiapa Yang Berkenaan

To Whom It May Concern

Dengan ini diakui bahawa

This Is To Certify That

5S2 / 2 / 2019

ARJUN THANNEERMALAI A/L SHANKER

telah belajar di Sekolah Menengah St. George, Taiping.

attended St. George's Institution, Taiping.

sejak

2015

hingga

2019

from

to

Dia telah memasuki Peperiksaan

SIJIL PELAJARAN MALAYSIA (SPM)

He/She sat for the

Nilai Akademik
Academic Qualities

PENCAPAIAN AKADEMIK AGAK MEMUASKAN . ADA USAHA UNTUK
MENCAPAI KEJAYAAN.

Jawatan bertanggungjawab dan Kegiatan luar darjah
Positions of Responsibility and Extra-Curricular Activities

1. DRAMA - PERINGKAT NEGERI 2018
2. INTERACT - AHLI AKTIF
3. ST JOHN AMBULANS - AHLI BIASA
4. KRIKET - AHLI BIASA

Perwatakan
Character

SEORANG PELAJAR YANG BERSOPAN SANTUN DAN JUJUR

Tarikh

31.12.2019

Date:



GOH BOON KUN

PENGETUA

SMK ST. GEORGE, TAIPING



SMK ST. GEORGE TAIPING, TAIPING
JALAN STESEN, TAIPING, 34000 LARUT MATANG & SELAMA, PERAK

Sijil Perakuan

UJIAN LISAN BERASASKAN SEKOLAH (ULBS)
SIJIL PELAJARAN MALAYSIA 2019

ARJUN THANNEERMALAI A/L SHANKER
021002020017
AG006A026

Calon di atas telah melaksanakan
Ujian Lisan bagi mata pelajaran

BAHASA INGGERIS

EN. GOH BOON KUN
PENGETUA
SMK ST. GEORGE TAIPING, TAIPING
18 OKTOBER 2019





SMK ST. GEORGE TAIPING, TAIPING
JALAN STESEN, TAIPING, 34000 LARUT MATANG & SELAMA, PERAK

Sijil Perakuan

UJIAN LISAN BERASASKAN SEKOLAH (ULBS)
SIJIL PELAJARAN MALAYSIA 2019

ARJUN THANNEERMALAI A/L SHANKER
021002020017
AG006A026

Calon di atas telah melaksanakan
Ujian Lisan bagi mata pelajaran

BAHASA MELAYU

EN. GOH BOON KUN
PENGETUA
SMK ST. GEORGE TAIPING, TAIPING
18 OKTOBER 2019



SEKOLAH MENENGAH KEBANGSAAN SAINT GEORGE
34000 TAIPING, PERAK DARUL RIDZUAN.



Calon yang tersebut nama yang di bawah telah menduduki

Pentaksiran Kemahiran Proses Sains dan dianugerahkan

PELAPORAN KEPUTUSAN

PENTAKSIRAN KEMAHIRAN SAINTIFIK

NAMA : ARJUN THANNEERMALAI A/L SHANKER

NO. KP : 021002020017

TAHUN : 2019

MATA PELAJARAN

MARKAH (%)

FIZIK : 75

KIMIA : 54

BIOLOGI : 75



GOH BOON KUN

Pengetua
SMK Saint George
34000 Taiping, Perak

Tarikh : 1 Dec 2019



PERSATUAN PENGAJIAN KITAB AGAMA HINDU MALAYSIA

Arsha Vidya Nilayam Malaysia

(Reg. No. 4175/96 Perak)

We are pleased to award this certificate to
Arjun Thaneermalai a/l Shanker

For his/her completion of a course under the Purna Vidya Programme
(Vedic Heritage Teaching Programme)

From 2013 to 2015 Level 1

Curriculum :

Purna Vidya Level 1

Epics : Ramayana and Mahabharata

Periapuranam: Stories of Naayanaars.

Prayers: Chanting of Selected Vedic Hymns, Stotras
and Thirumurai Hymns,

Worship: 16 step pujavidanam

Guided Meditation.

Moral value education: Universal values and Thirukural.

Bhajans: Singing of devotional songs.

Yoga: Basic Yoga Training.

Purna Vidya Teachers

G. Sathasivam

G. Sathasivam

M. Rajamane

M. Rajamane

Jitatmananda

Swami Jitatmananda Saraswati
President & Acharya in charge

President
ARSHA VIDYA NILAYAM



(RPN: 23663)

LEMBAGA PEPERIKSAAN
KEMENTERIAN PENDIDIKAN MALAYSIA

(LP/KOM.6) Pin. 1/2013

SIJIL PELAJARAN MALAYSIA 2019

NAMA : ARJUN THANNEERMALAI A/L SHANKER
NO. K/P : 021002-02-0017
SEKOLAH : SMK ST GEORGE TAIPING, TAIPING

ANGKA GILIRAN : AG006A026
KOD DAFTAR : 8 JEN.DAHULU: 0

KOD	NAMA MATA PELAJARAN	GRED
1103	BAHASA MELAYU	C KEPUJIAN
1119	BAHASA INGGERIS	A CEMERLANG TINGGI
1225	PENDIDIKAN MORAL	B+ KEPUJIAN TERTINGGI
1249	SEJARAH	C KEPUJIAN
1449	MATHEMATICS	B+ KEPUJIAN TERTINGGI
3472	ADDITIONAL MATHEMATICS	E LULUS
4531	PHYSICS	D LULUS ATAS
4541	CHEMISTRY	E LULUS
4551	BIOLOGY	E LULUS

1119(GCE-O) - 3B

LAYAK MENDAPAT SIJIL

PENGARAH PEPERIKSAAN



KEMENTERIAN PENDIDIKAN MALAYSIA
JABATAN PENDIDIKAN NEGERI PERAK

Sijil Pencapaian

Setinggi-tinggi penghargaan dan tahniah kepada

ARIUN THANEERMALAI A/L SHANKER
021002-02-0017

Atas pencapaian cemerlang

JOHAN

PERTANDINGAN DRAMA BAHASA INGGERIS
SEKOLAH-SEKOLAH MENENGAH PERINGKAT NEGERI PERAK
TAHUN 2018

Bertempat di

SMK SUNGAI KRUIT, SUNGKAI

Pada

24 JULAI 2018



05028

PerakE xcellent

DATO' HAJI MAT LAZIM BIN JORIS, DPMP., AMP.
PENGARAH PENDIDIKAN NEGERI PERAK



KEMENTERIAN
PENDIDIKAN
MALAYSIA

Pejabat Pendidikan Daerah
Larut Matang dan Selama

Sijil Pencapaian

Dianugerahkan kepada

ARJUN THANEERMALAI A/L SHANKER

021002-02-0017

JOHAN

**PERTANDINGAN DRAMA BAHASA INGGERIS SEKOLAH MENENGAH
PERINGKAT DAERAH
LARUT MATANG DAN SELAMA
TAHUN 2018**

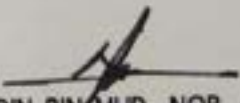
Pada

18 JULAI 2018

Bertempat di

SMK CONVENT, TAIPING




HAJI AMINUDIN BIN MHD. NOR, AMP., PPT.
Pegawai Pendidikan Daerah
Larut Matang Dan Selama

SEKOLAH MENENGAH KEBANGSAAN SAINT GEORGE
34000 TAIPING, PERAK DARUL RIDZUAN.



Anugerah

KECEMERLANGAN KOKURIKULUM

ARJUN THANNEERMALAI A/L SHANKER

021002020017

atas pencapaian

JOHAN

PERTANDINGAN DRAMA BAHASA INGGERIS
PERINGKAT DAERAH LMS DAN NEGERI PERAK

TAHUN **2018**



GOH BOON KUN

Pengetua
SMK Saint George
34000 Taiping, Perak

Tarikh : 27.10.2018



KEMENTERIAN
PENDIDIKAN
MALAYSIA

Certificate of Participation

This certifies that

ARJUN THANEERMALAI A/L SHANKER
021002-02-0017

has participated in the

NATIONAL ENGLISH LANGUAGE DRAMA COMPETITION

held on 22nd – 25th October 2018,
at IPG Tun Hussien Onn, Batu Pahat, Johor

ROZAINUM BINTI AHMAD K.M.N.
DIRECTOR
CO-CURRICULAR AND ARTS DIVISION,
MINISTRY OF EDUCATION MALAYSIA



Nombor Siri :
26747



Koperasi
Tenaga Muda
Taiping



SURUHANJAYA
KOPERASI
Malaysia



KEMENTERIAN
PENDIDIKAN
MALAYSIA

Pusat Pendidikan Daerah Larut Matang dan Selama

SURUHANJAYA KOPERASI MALAYSIA
NEGERI PERAK

Sijil Penyertaan

Dengan ini disahkan bahawa

No. Kad Pengenalan

Telah menyertai dan tamat dengan jayanya

**LARIAN
KOPERASI
PERAK
2019**

yang telah diadakan di
PADANG ESPLANADE, TAIPING

Pada

15 SEPTEMBER 2019

LT. KOL. SYED NASIR BIN SAYED MAHADI
Pegawai Pendidikan Daerah
Larut Matang dan Selama

AZAM BIN SULAIMAN
Pengarah
Suruhanjaya Koperasi Malaysia
Negeri Perak



Sijil Penyertaan

Adalah diperakui bahawa

ARJUN SHANKER
021002-02-0017

telah mengambil bahagian
dalam

Pertandingan Kaligrafi Cina

sempena sambutan Perayaan Tahun Baru Cina

anjuran PERSATUAN BAHASA CINA

SMK ST.GEORGE

pada

31 Jan 2019



GOH BOON KUN
PENGETUA
SMK ST.GEORGE